Demonstration of Proposed Features

Date	02 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

The proposed features of the OptiCrop project were successfully implemented and demonstrated. The application accepts soil nutrient and environmental parameters, processes them using a trained Machine Learning model, and recommends the most suitable crop. The project also includes data preprocessing, exploratory data analysis, model training, and a responsive Flask web interface. The demonstration verified that all implemented features function correctly and meet the project's objectives.

Proposed Features :

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation System	Predicts the most suitable crop using soil nutrients and environmental parameters.	Implemented	Yes	Successfully demonstrated using sample inputs.
2	Data Preprocessing	Handles missing values, outlier removal, and data preparation before model training.	Implemented	Yes	Preprocessing workflow demonstrated successfully.
3	Exploratory Data Analysis (EDA)	Performs statistical analysis and visualizes relationships within the dataset.	Implemented	Yes	Charts and analysis results were presented
4	Machine Learning Model Training	Trains and evaluates machine learning models for crop prediction.	implemented	Yes	Model accuracy and evaluation metrics demonstrated.
5	Flask Web Application	Provides an interactive interface for entering parameters and viewing crop recommendations.	Implemented	Yes	Web application demonstrated successfully.
6	Input Validation	Validates user inputs on both client-side and server-side before prediction.	Implemented	Yes	Invalid inputs were handled correctly during demonstration.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%